AI-MEDIA AUTHENTICITY SCREENING

AI-Media Guard

A local tool for estimating whether images and videos look AI-generated

REAL / AI?

an estimate, not proof

IMAGE + VIDEO

PROJECT PRESENTATION · COMPUTER VISION

The authenticity problem

Realistic synthetic media can be difficult to judge by sight alone.

People need a simple first check when an image or video feels uncertain.

PROJECT GOAL

Analyze uploaded media and return a likely class with a confidence score.

The result supports a human judgment. It does not prove who created the media or whether it has been altered.

System overview

01

Upload media

JPG, PNG, WEBP, MP4 or MOV
Maximum upload: 25 MB

02

Analyze content

One image is scored directly.
Video is sampled into frames.

03

Show estimate

Likely REAL or AI-generated with a confidence score

Frontend and API run together on the same local server.

Model and training

MODEL

EfficientNetB0

ImageNet pretrained feature extractor;
base layers frozen during this training run.

A small dense head learns the FAKE /
REAL decision.

TRAINING SETUP

CIFAKE dataset
12,000 training images per run
128 × 128 input size
3 epochs, CPU training
5,000 held-out test images

Output: one probability for REAL; the complementary score represents FAKE.

Video analysis uses sampled frames

The image model is reused for short videos.

Video file	Up to 8 frames	Image model	Average scores
MP4 / MOV	sampled across the clip	scores each frame independently	one final estimate

This is frame aggregation, not a temporal deepfake detector.

Initial evaluation

HELD-OUT CIFAKE TEST SUBSET

85.6%

test accuracy

5,000 test images sampled from the dataset

0.932

ROC AUC

test AUC = 0.9321

Early result on CIFAKE; performance on other media may differ.

Limits and next steps

CURRENT LIMITS

- CIFAKE covers a specific data distribution
- Video scores average sampled frames
- No separate manipulated-media class
- Confidence is not proof of provenance

NEXT STEPS

Evaluate on external datasets

Train a video-specific model

Report precision, recall and F1

Test images from newer generators

Use the result as a screening signal, then verify important claims independently.

Live demo

Start the local API

```
uvicorn app:app --reload
```

Open in a browser

http://127.0.0.1:8000

Upload → Analyze → Review

Try one CIFAKE image first. Then try a short MP4 or MOV clip and compare the confidence score.

Questions?